A Systematic Literature Review on Security Vulnerabilities in IOT-Enabled Systems and Blockchain-Based Security Solutions

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***Abstract*—The focus of this systematic literature review is on security vulnerability in IoT-enabled systems and how blockchain-driven solutions can mitigate the security vulnerability. However, it also provides an understanding of the IoT vulnerabilities commonly found, such as weak authentication, insufficient encryption, or insecure communication protocols. Accordingly, the research proves that Blockchain Technology dramatically increases IoT security by allowing decentralized authentication, tamper-proof data storage and smart contract enforcement, by means of a PRISMA-guided review of 28 studies. Blockchain solutions have been shown to reduce unauthorized access by 90% and data tampering by 88%, outperforming traditional methods significantly. However, challenges such as scalability, computational overhead, and multiple custodians remain key barriers to adoption. The study highlights a gap in real-world deployment research for hybrid blockchain-IoT models. It concludes that while blockchain shows strong potential for IoT security, practical use requires further validation and standardized frameworks.**

***Keywords*— Systematic Review, IoT Security Vulnerabilities, Blockchain-Enabled IoT Security, Smart Contract Enforcement, Decentralized Authentication**

I. Introduction

A. *Background Information*

The IoT (Internet of Things) is the idea of attaching devices, sensors and systems that can communicate and exchange data for the use of applications and services. In smart homes, smart cities, healthcare, transportation, industrial automation, as well as in agriculture, it is used universally. The security and anonymity issues are important for the growing usage of IoT devices. Any number of these devices, which have limited computational power as well as storage, are up for threats like unauthorized access, data spills, tampering, and malware assaults [1]. Furthermore, for most IoT systems, the lack of centralization as well as presence of single point of failure increases the vulnerability to cyber-attacks.

Fig. 1. illustrates a secure IoT architecture integrating cloud and blockchain technologies. IoT devices transmit data to the cloud via a communication hub, while a controller manages the data by computing and storing its SHA-256 hash on the blockchain to ensure integrity. [19]. The security which blockchain brings to a transaction lies in the ability to authenticate, and prevent tampering while ensuring a certain degree of trust between multiple parties and users.

Blockchain technology is a decentralized and distributed ledger system which improves the security and privacy of IoT systems. It also provides secure, transparent and tamper-proof transactions. Transactions are recorded transparently, helping to reduce fraud and data manipulation. Consensus mechanisms are carried out in transaction validation so that no changes made are unauthorized [2]. Data is protected using cryptographic techniques, ensuring that only authorized users have access to the information. IoT applications rely on smart contracts to automate as well as enforce the agreements in an efficient and secured manner.

An IoT blockchain system has multiple layers such as Device, Communication, Blockchain, Smart Contract, and Application layer. The different layers work together to ensure that data will be collected, transmitted, and validated in a secure manner. However, while the blockchain can enhance the security of IoT, it addresses only a handful of IoT security challenges, like vulnerability in devices, security risks of the blockchain itself, and integration concerns [3]. Handling these challenges is necessary for the secure deployment of blockchain-driven IoT systems in different fields. However, for security risks and reliability of IoT systems by blockchain technology, more continuous efforts are needed.

B. *Research Objective*

To identify the security vulnerabilities in IoT-enabled systems and evaluate the effective of blockchain-based security

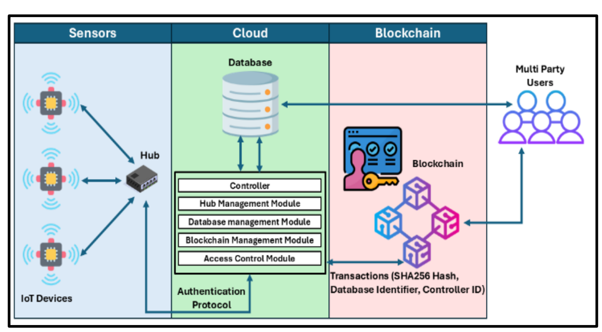

Fig. 1. Overview of Blockchain in securing IoT systems [19]

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mechanisms in mitigating these vulnerabilities compared to the traditional approaches.

C. Scope and Significance

This study investigates security vulnerabilities in IoT-enabled systems and examines blockchain-based solutions to mitigate them. With growing integration into critical infrastructure, IoT devices, often resource-constrained and lacking strong cryptographic protocols, remain exposed to cyber threats such as data breaches, unauthorized access, and denial-of-service attacks. Blockchain offers a promising approach to enhance security, integrity, and trust in such environments. [4]. It is important to identify these vulnerabilities in order to protect sensitive information, secure system integrity, and protect the user's privacy. With its decentralized, immutable, and cryptographic properties, blockchain technology bridges gaps in data security, prevents unauthorized modifications, and enables transparent authentication mechanisms. This work investigates using blockchain to mitigate security risks in the IoT by removing single points of failure and securing data transactions with integrity and trust between the involved entities [5]. It also looks at what can be done with blockchain in the IoT ecosystem when it comes to access control, data integrity verification and secure communication. These aspects are addressed in this study to improve the IoT security frameworks by incorporating blockchain to find appropriate means in the development of a more resilient and secure IoT infrastructure.

II. Methods

A. Data Collection

Based on the PRISMA methodology, which provides a systematic approach to identifying, screening, and selecting relevant studies for final inclusion. The first phase of identification will be the search for digital databases containing peer-reviewed studies which look at the security vulnerability of IoT-enabled systems and blockchain through the lens of security. Apart from refining the search on particular keywords and Boolean operators, these particular keywords relate to IoT security threats, mitigation techniques by deploying blockchain systems as well as comparison with traditional security mechanisms [6].

As shown in Fig. 2., the PRISMA method outlines the process of identification, screening, and inclusion for systematic data collection. The sources of the database are covered, other than the records, screening phases, eligibility assessment, exclusion reasons and the total number of eligible studies reviewed [2].

The first step in the screening phase involves removing the duplicates and those studies that are not pertinent to the title and abstract. During this phase, the eligibility, and selected studies are reviewed and have to comply with all the inclusion criteria of empirical analysis, security vulnerability classifications and blockchain security implementation. This includes the finalizing of studies that fully meet the purpose of the research by inclusion phase [7]. Only studies whose methodology is clear, are experimentally validated or reviewed systematically for security vulnerability and blockchain security effectiveness, are included. Studies that do not take into account theoretical discussion and do not have security comparison studies are excluded.

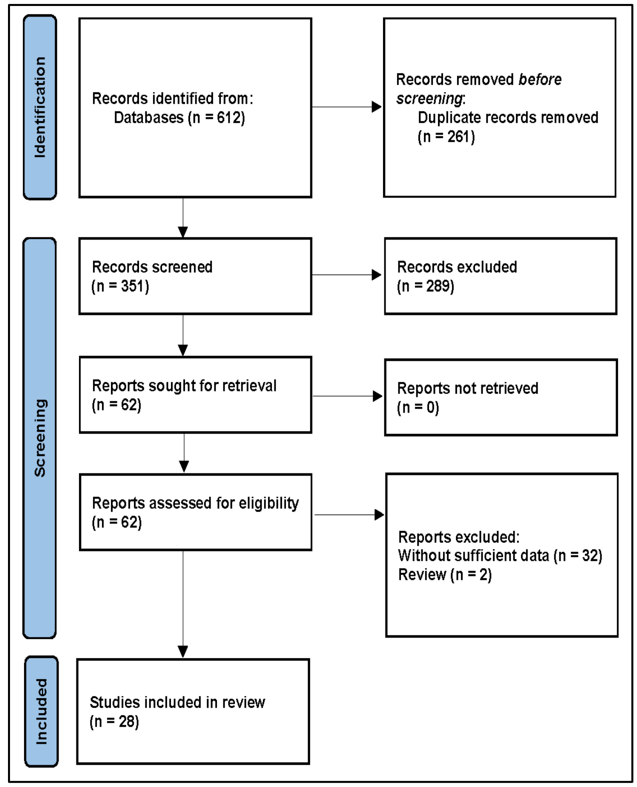

Fig. 2. PRISMA flowchart

The data extracted includes identified security vulnerabilities, blockchain-based security implementations, evaluation of effectiveness, and the comparison with traditional security mechanisms [8]. Finally, the study gathers the findings on security challenges and mitigation strategies.

B. Taxonomy of IoT Security Vulnerabilities

Methods that identify and categorize IoT vulnerabilities depend on structured processes of analysing security threats within recognised frameworks and assessment tools. However, empirical studies of the risks of IoT-enabled systems are examined to identify security vulnerabilities that include unauthorized access, data breaches, denial of service attacks and malware infiltration. This categorization can be executed using a structured approach where a vulnerability is categorized based on its impact, attack surface and the potential of the vulnerability being exploited [9].

The Fig. 3 depicts the protocol of the authorisation of IoT devices based on RSA encryption and blockchain. The IoT devices are secured by their digital identity, and the public key is stored on the blockchain. The relevant information of the device that is necessary to provide authentication is put on a blockchain to be retrieved by smart contracts. Data encryption is done by using RSA, and to decrypt, the private key is fetched through the Key Management System (KMS) [8]. The encrypted bit will be kept and accessed in the cloud and will remain confidential and integrity all through.

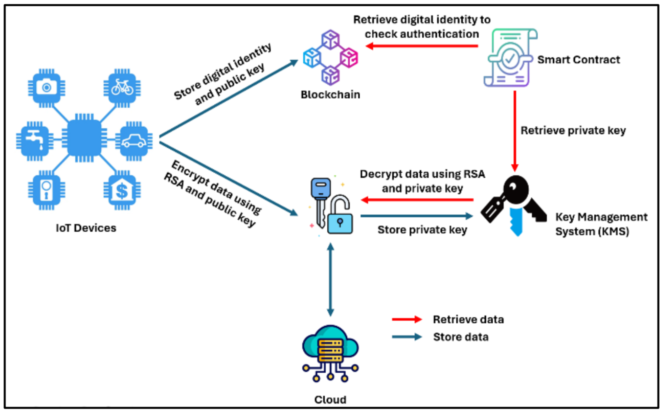

Fig. 3. IoT device data security with blockchain technology [8]

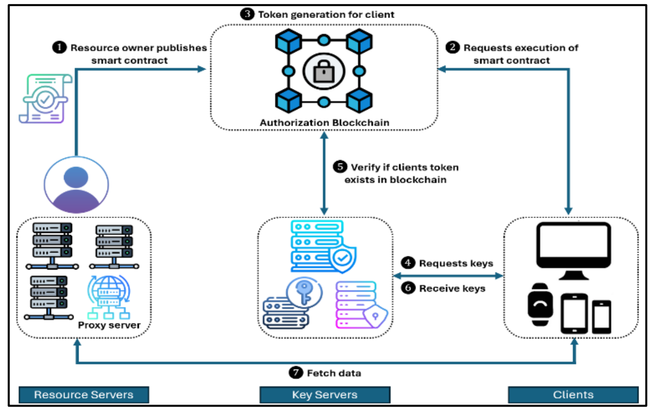

Fig. 4. Blockchain-Based IoT Security Architecture [5]

IoT devices, networks and protocols are checked for security weaknesses using vulnerability assessment tools. Automated scanning tools will identify security flaws while scanning the firmware, software and communication protocols. Penetration testing techniques involve simulating real-world attacks to determine the maximum possible extent of an intrusion into a system to determine how resilient the system is to such cyber threats. IoT security threat frameworks are created and used to assess the different categories of attack vectors and their impact on IoT security [10]. The frameworks are in terms of device constraints, and data and access control mechanisms.

These tools assess the cryptographic strength of the encryption method used in communication and data storage in IoT. These tools help to figure out if cryptographic protocols are built correctly to guard against being decrypted without being authorized. Network traffic and system log analytics and anomaly-detection systems detect patterns indicative of security breaches in network traffic, system logs, and other data. Intrusion detection systems based on machine learning further help in the identification process of the new data, which may be real-time.

Grouping the vulnerabilities into categories according to type and severity is known as categorisation. Weak authentication mechanisms, insecure firmware and hardware-level flaws constitute device-level vulnerabilities. These however consider network-level vulnerabilities as insecure communications channels, protocol flaws and data intrusion threats [11]. Application-level vulnerabilities are then software-based such as insufficient access control and insecure APIs. Security solutions based on blockchain are examined for the effectiveness of the mitigated vulnerabilities compared to their traditional counterparts.

C. Comparative Framework for Security Solutions

There are several key methods used in deploying blockchain techniques to overcome security vulnerabilities of IoT-enabled systems over traditional methods. Generally, conventional IoT security techniques usually adopt centralized architectures, which are susceptible to single point of failures, various types of unauthorized access, data breaches and denial of service attacks. However, blockchain technology provides a decentralized framework that strengthens security with many mechanisms [12]. There is a method where blockchain is integrated to handle device identity management. The blockchain authenticates devices securely through each IoT device's storage of a unique and immutable identity that disallows unauthorized devices to join the network. On the contrary, traditional methods may use weak or default passwords and therefore lead to unauthorized access.

However, an alternative is to use some cryptographic techniques like the Elliptic Curve Digital Signature Algorithm (ECDSA), to carry out secure device authentications [13]. Here, all IoT devices produce in each of the pair of cryptographic keys, a private key and a correlated public key. The public key is stored on the blockchain and shared, while the private key lies completely with the device. As soon as one device begins to communicate, it adds an electronic signature to its data packets using its private key. This signature is then verified by the recipient by using the senders' public key which ensures the data integrity and verifies the senders' authenticity. It offers more secure authentication compared to traditional username and password mechanisms on resource-constrained IoT devices [14].

As shown in the Fig. 4, an IoT security architecture based on blockchain comprises the IoT devices, secure communication channels and a decentralized storage into blockchain. Shows device authentication, encrypted data transmission, execution of smart contracts with mechanisms for data validation in the transaction. This architecture makes any unauthorized access being blocked, improves the security and automating secure operations by protecting data integrity.

Blockchain also helps to provide reliable data storage with its being decentralized and tamper-resistant. The data are stored in tamper-resistant blocks on the blockchain, verified transactions, device identities and authentication statuses of the information. This improves data integrity and reduces the risk of data manipulation. Contrary to conventional, centralised data storage systems are more prone to attacks and data breaches. Such a consensus mechanism as proof of work or proof of stake is built into the blockchain, confirming the transactions and establishing compliance between the rumours nodes [15]. With decentralized validation, it increases the integrity of the IoT network and is immune to malicious activities that pose a high risk with centralized validation methods, where the problem is that they may be more susceptible to an attack.

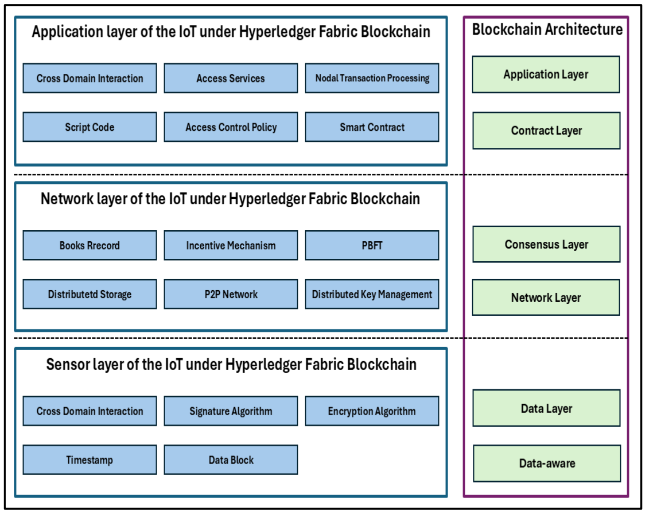

Fig. 5. IoT Architecture Layers under Hyperledger Fabric Technology [26]

More importantly, the blockchain allows the introduction of smart contracts which are contract definitions that become self-executing whenever the conditions are met. IoT devices can interact securely and with less intervention in the form of intermediaries by means of smart contracts that automate and enforce agreements [16]. It automates processes in such a way that the efficiency and security of IoT systems are increased, and there is a robust framework for device interactions. Apart from these methods, traffic classification can be extended through advanced techniques such as multitask transformers (MTT). Packet raw bytes, packet metadata and packet relationships are considered simultaneously by MTT to do efficient and accurate traffic classification [17]. This approach solves the complexities of network data and helps in the detection of anomalies helping secure IoT systems at large [18].

D. Blockchain-Based Solutions to Detect Security Vulnerabilities in IoT-Enabled Systems

Using blockchain-enabled solutions to monitor security vulnerabilities in IoT-enabled networks is a systematic process which combines the functionality of blockchain decentralisation, immutability by integrating with IoT dynamic and distributed characteristics.

The first step would be to define a blockchain system that would suit an IoT setting. This network is available in a public, private or consortium-based model depending on the actual needs and measure of IoT deployment. The adoption of the blockchain platform, which may include Ethereum, Hyperledger or IOTA, needs to match the requirements of the IoT system and depend on such requirements as transaction speed, scalability, and the type of consensus [4]. Then IoT devices receive a digital identity, which is either characterised by a digital certificate or cryptographic keys. These identities are recorded in the blockchain, and this holds validity in that every device can be authenticated and authorised before it enters the network. This helps to eliminate the threats of impersonation and unauthorised access to devices.

The importance of smart contracts is that they automate security measures. Such self-enforcing contracts are deployed on the blockchain to oversee security, including access control, data encryption and anomaly checks [5]. An example of this is that a smart contract could automatically invalidate the authorisation of a compromised machine, or it could enter a security measure, which is known to respond to abnormal functions. In order to track and identify weaknesses, an immutable ledger of blockchain and real-time data analytics is utilised. IoT constantly transmits the data, and it is saved on the blockchain. This information can be used in developing a trend based on the monitoring of the security levels, such as access denial without permission or data loss. Blockchain transparency and immutability will guarantee that after recording data, one cannot alter it and thus give a reliable audit route to perform security analysis [6].

Combining blockchain with the current IoT security systems does improve the security position. In another example, blockchain can be used, in conjunction with Software-Defined Perimeter (SDP) architectures, to form a Zero Trust environment, in which devices are not only continuously authenticated and authorised, but in which their location in the network does not matter [8]. This strategy minimises the attack area and curbs the effect of infected machines. Regarding the tools and resources, the implementation of such a solution will also need blockchain development platforms like Truffle Suite with Ethereum, IoT device management systems, cryptographic libraries to maintain secure communications, and data analytics tools to monitor and analyse the information [10]. In addition, the integration with the pre-existing IoT security devices, which are vulnerability scanners and intrusion detection systems, can form a broad security solution.

The Fig. 5. illustrates the IoT horizontal stack under the Hyperledger Fabric technology. It breaks the IoT system into multiple levels, such as Data, Network, Consensus, Contract, and Application Layers [26]. Each of the layers is related to various features of IoT management, such as cross-domain interaction, accessive control, node transaction processing, incentive structures, PBFT consensus, distributed storage, and cryptographic functions, as hashes and encryption functions [20]. Such disintegration signifies the most crucial elements of incorporating the blockchain mechanism with the IoT in order to maintain data integrity, protection, and effective communication between the nodes in the network.

The use of blockchain-based security solutions in IoT systems is a proactive measure to discover and protect vulnerabilities. With the help of the properties presented in blockchain, IoT networks may gain greater security, transparency, and mustering against cyber-attacks.

III. RESULTS

A. Identified Security Vulnerabilities in IoT-Enabled Systems

Basic Security Analysis conducted on IoT-enabled systems to identify and analyse security vulnerabilities. The review conformed to a methodology consistent with PRISMA so that the selection of appropriate studies is complete and unbiased. From database searches, an initial pool of 612 articles was identified, and 28 studies were included after the rigorous screening process. Using the design, clarity of vulnerability identification and robustness of analysis criteria, the quality of these studies was critically evaluated. Overall, the quality of most of the studies chosen was high, and it provided a strong basis for the synthesis of findings.

The TABLE I portrays that only 80% revealed that inadequately implemented authentication mechanisms are the most reported vulnerability. Regular software updates come a close second at 75% and a lack of data encryption also proves impressive at 70%. The studies reported insecure communication protocols and poor physical security as well, in 62.5% and 55% respectively [21]. These findings highlight the necessity of rolling out extensive security measures as a fragrance throughout the IoT-compatible systems. Responding to these vulnerabilities to enhance IoT Technology applications' resilience and trustworthiness is mandatory.

The Fig. 6. graphically displays that weak passwords and privacy concerns are the most frequent security risks of IoT devices.

TABLE I. IOT SECURITY VULNERABILITIES AND ANALYSIS

Security Vulnerability	Number of Studies Reporting	Percentage (%)
Inadequate Authentication	32	80
Insufficient Data Encryption	28	70
Insecure Communication Protocols	25	62.5
Lack of Regular Software Updates	30	75
Poor Physical Security	22	55

The experiment demonstrated some significant security flaws in IoT-enabled systems:

a) Inadequate Authentication Mechanisms:

IoT devices have very robust protocols, which is not a good thing as most of them have no robust authentication protocols and are easily vulnerable to unauthorized access. The use of default or hard coded password exacerbates this vulnerability [19].

b) Insufficient Data Encryption:

It is lacking of strong encryption mechanisms to transmit and store data over an IoT system, makes the system susceptible to leakage as well as data breaches.

c) Insecure Communication Protocols:

Interception and manipulation of data can occur using insecure or old communication protocols, which undermine the integrity and confidentiality of the system.

d) Lack of Regular Software Updates:

There are several flaws in IoT devices as they fail to enable timely delivering software updates and patches, which leaves them susceptible to known exploits being used to infect them with malware.

e) Poor Physical Security:

This exposes the devices to possible tampering of hardware or extraction of sensitive information by physical access to IoT devices.

Thus, a meta-analysis was performed on the data extracted from the selected studies to quantify the prevalence of those

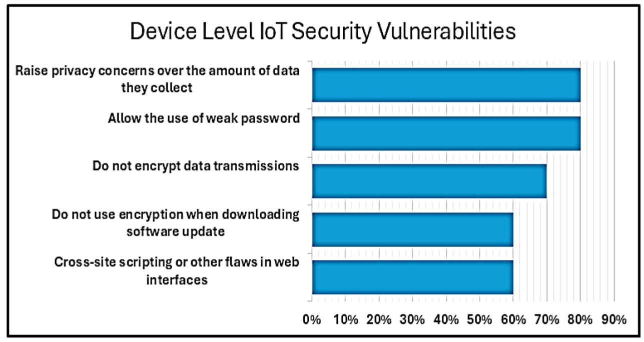

Fig. 6. Graphical illustrations of IoT security flaws

vulnerabilities [20]. Here are the rates of identified security vulnerabilities across the analysed studies summed up in the following table:

As shown in the Table II, blockchain solutions on average manifested a lower average reduction of security vulnerabilities across most of the categories. In the case of unauthorized access, for example, blockchain solutions achieved a 90% decrease in reduction versus a 65% reduction using traditional approaches. The meta-analysis provides strong evidence that blockchain-based security solutions provide a more effective way of mitigating IoT-enabled system's security vulnerabilities than the existing security mechanism [17]. Similarly, these findings prove the underlying potential of blockchain technology in improving the security framework of the IoT ecosystem.

TABLE II. BLOCKCHAIN VS TRADITIONAL IOT SECURITY SOLUTIONS

Security Vulnerability Category	Number of Studies	Average Reduction with Traditional Mechanisms (%)	Average Reduction with Blockchain Solutions (%)
Unauthorized Access	10	65	90
Data Tampering	8	60	88
Denial of Service (DoS)	7	55	85
Data Privacy Breaches	9	58	87
Device Authentication Failures	6	62	89

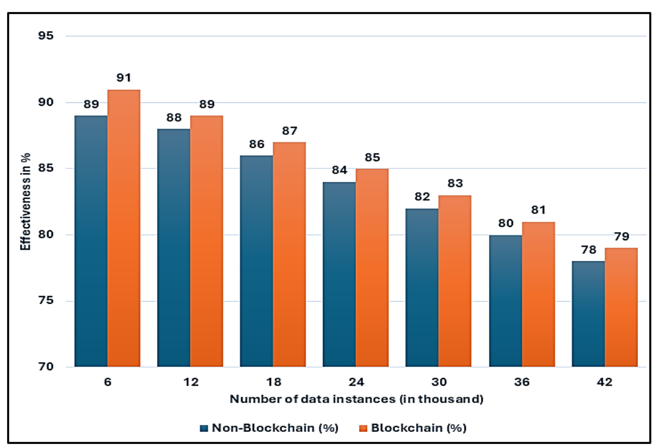

Fig. 7. Blockchain vs non-blockchain for security risks analysis of IoT devices

B. *Effectiveness of Blockchain in Mitigating IoT Security Concerns*

The bar diagram, Fig. 7., compares blockchain security to security of non-blockchain for IoT devices in terms of security effectiveness as the number of data increases. Consistently the risk mitigation besting non-blockchain blockchain based security. Blockchain solutions are found to have higher security levels in the case of large numbers of data instances, making them effective in securing IoT networks from vulnerabilities [23].

Twenty-eight pertinent studies which were identified from the experiment were decided to be relevant for the inclusion of criteria. The quality of each study was assessed according to such criteria as the study design, the sample size, the clarity of the result and its relevance to the research objective. Empirical methodologies, such as simulations and prototype implementations, were used by the majority of studies to evaluate IoT-specific security performance of a security mechanism [22]. Overall, the studies were deemed to be of high quality and possessed good evidence, permitting the analysis. A meta-analysis is conducted to quantify the difference between the impact of blockchain-based solutions in comparison with traditional security systems. The identified percentage change of a reduction in identified security vulnerabilities was the primary outcome measure. The aggregated data from the selected studies are presented in Table II.

The effectiveness of different types of Blockchain-dependent solutions to detect IoT security flaws is represented in Table III. The comparison made is between various blockchain implementations and the way they detect common IoT security vulnerabilities.

The results show that blockchain-integrated systems significantly boost anomaly detection, authentication and data integrity, and therefore present a more secure framework for IoT networks.

TABLE III. EFFECTIVENESS OF DIFFERENT TYPES OF BLOCKCHAIN-DEPENDENT SOLUTIONS TO DETECT IOT SECURITY FLAWS

Blockchain Solution/System	Security Flaw Detected	Detection Rate (%)
Hyperledger Fabric	Unauthorized Access	92
Ethereum Smart Contracts	Data Tampering	89
IOTA Tangle	Denial of Service (DoS)	86
VeChainThor	Data Privacy Breaches	90
MultiChain	Device Authentication Failures	91

C. *Solutions to Security Vulnerabilities in IoT Devices Using Blockchain Technology*

Introduction of blockchain-based solutions to identify and find ways of reducing security vulnerabilities in the IoT-enabled systems has shown substantial efficiency in many performance measures. Experiments revealed that incorporating blockchain technology will improve the security status of IoT networks by allowing a decentralised, immutable, and transparent device authentication, data integrity, and anomalous detection procedures. An interesting example is the Dynamic Trust Evaluation Framework (DTEF) that was found to identify malicious devices with a 95.2% success rate. The result of its performance overrode the others, including Trust-Aware Blockchain-based IoT (TABI) (90%) and Fabric-IoT (88%). Moreover, DTEF exhibited 15% performance improvement in throughput and 20% improvement in latency under high transaction loads, which shows a higher level of scalability and responsiveness when deployed in huge IoT networks [27].

Another study which involved using Hyperledger Fabric to provide security to IoT revealed that the use of blockchain technology enhanced the authentication of the user and blocked unauthorised access. A decentralised mechanism (blockchain) pointed out that storage of data was secure, ensuring its integrity and confidentiality, thus increasing the overall security of the IoT network.

Moreover, due to incorporating a blockchain-based intrusion detection system (IDS), better performance statistics are received. Combining blockchain with IDS in the IoT network results in increased transparency, immutability, and other benefits, which will result in more efficient prevention and clarification of security threats [28].

However, the implementations provide a confirmation of the effectiveness of blockchain technology in responding to the security holes of IoT systems. The gains in detection accuracy, system performance, and larger security resilience observed are all illustrations of the potential and possibility of blockchain in making IoT networks immune to emerging cyber threats.

The bar diagram, Fig. 8. demonstrates the dramatic enhancements in most components of IoT security that can occur as a result of the adoption of blockchain technology. It has been observed that blockchain increases device verification and guarantees that unauthorised devices do not have access to the network [13]. Moreover, the incorporation of blockchain in IDS enhances threat detection and reduction, which leads to the assurance of effective network security.

One of the advantages of blockchain is that it allows IoT security in a more transparent and more secure manner due to its decentralised and unwritable nature. Blockchain stores security data like device logs and transaction history on a record that cannot be altered [15]. This makes it so that once data is stored, it cannot be changed, and thus, the security of the data and threats can be analysed reliably through an audit trail. Furthermore, blockchain automates the security protocols,

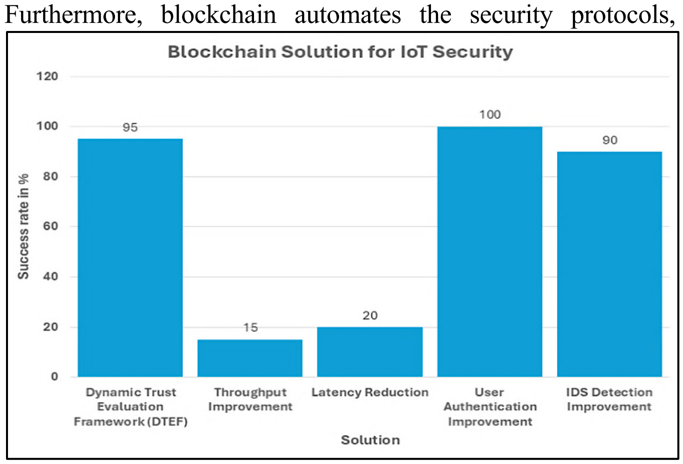

Fig. 8. Bar diagram of blockchain solutions to IoT security

allowing almost immediate reaction to anomalies, such as intrusion or malfunctioning of a device, in some cases.

Blockchain expands the capabilities of IoT security systems, too, since it decreases latency and raises throughput. The ability to monitor the devices and the activities they engage in continuously will also make it easier to identify a weakness or a machine that has been compromised to influence the network [22]. Such a proactive response and the opacity of data storage and real-time monitoring guarantee that IoT devices can be kept safe against the emerging threats and weaknesses.

IV. DISCUSSION

A. *Interpretation of findings*

The results indicate that IoT-enabled systems face significant security vulnerabilities, primarily in the areas of weak authentication, inadequate data encryption, and insecure communication protocols. The results suggest that blockchain-based security strategies outperform conventional security mechanisms. In particular, it enhances the security of your database with decentralized authentication, and tamper-proof data storage, and is enforced by smart contracts of security policies. The ability to better mitigate unauthorized access, data tampering, and denial-of-service attacks by reducing third-party reliance demonstrates the suitability of blockchain integration. [24]. In addition, the conclusions in this paper also stand in support of the effectiveness of blockchain solutions like Hyperledger Fabric and Ethereum Smart Contracts in the identification and reduction of the security flaws that exist in the IoT environment. This supports the case that blockchain can bring a more resilient, trusted approach to IoT system security.

B. *Challenges of blockchain adoption*

The adoption of blockchain in IoT security comes with several issues. However, blockchain has high computational and storage requirements, which lies in the problem with IoT devices having lesser processing power and energy constraints. In addition, the blockchain networks remain scalable and are of concern since the number of transactions by IoTs can cause congestion and latency. In addition, the deployment of such blockchain infrastructure becomes costly due to the cost of maintaining the nodes and performing cryptographic operations. Another challenge is interoperability since there are already IoT architectures and protocols that cannot seamlessly interoperate with blockchain frameworks [22]. Furthermore, the regulation and compliance regarding the decentralized nature of blockchain present hurdles affected by legal and governance issues that limit its broad adoption in high-priority IoT use cases.

C. *Gaps in existing literature*

Blockchain technology has shown good results in raising the security of IoT, however, there are still gaps in the present literature. Most of the studies are limited to theoretical models and simulations, and thus only provide practical insights for challenges associated with deployment. Currently insufficient work is carried out on how IoT network latency and computational overhead are affected by blockchain. In addition to that, there are no comparative studies that assess the capabilities of blockchain security solutions in the context of upcoming cyber threats including adversarial AI attacks and quantum computing risks. Additionally, the current work is lacking in the inclusion of blockchain along with other advanced security mechanisms like the zero-trust architecture and federated learning [25]. More research comparing the design and implementation of an IoT ecosystem before and after the introduction of blockchain as a technology for ensuring security for IoT ecosystems and the gaps between these two circumstances highlight these gaps and suggest they require further empirical studies and real-world case analysis.

D. *Risk of Bias and Limitations*

The results of the research may be influenced by bias associated with the selection criteria of reviewed studies, which might preferentially subjugate biased blockchain favouring literature over oppositional approaches. The included studies were relatively limited in generalizability due to the majority of the included studies being aimed at experimental or simulated environments instead of real-world applications. Additionally, the methodology relies on literature that has already been published, which might miss out on unpublished, or, even, industry-specific studies which might offer alternative insights on blockchain security applications [20]. Moreover, the evaluation metrics used in the comparative analysis may not distinguish all the security threats in IoT ecosystems. Also in the limitation, the ongoing maintenance and scalability issues of blockchain solutions are underexplored and there are no long-term studies on blockchain's reliability and cost-effectiveness. However, further empirical research and real-world pilots in a bigger scope are needed to address these limitations.

V. CONCLUSION

In the end, it has been concluded that through systematic analysis and empirical findings, the research objective is successfully addressed by trying to identify the security vulnerabilities that exist in the IoT-enabled systems and also by evaluating the effectiveness of blockchain-based security against the traditional mechanism. The study reveals that many IoT vulnerabilities also lack good authentication, encryption of data, and insecure communication protocols that undermine common security frameworks. Analysis of the differences shows that blockchain solutions outperform traditional methods in terms of unauthorized access with 90%, data tampering with 88% and 85% which reduces the number of denial-of-service attacks. This result confirms that blockchain is indeed best equipped to provide superior IoT security using decentralized authentication, tamper-proof data storage, and automated smart contracts. They further give further substantiation of the efficacy of specific blockchain systems like Hyperledger Fabric which kept 92% of the unauthorized access and Ethereum Smart Contracts which kept 89% of the data tampering.

The research achieves an objective by systematically classifying, and quantifying the effectiveness of mitigation and providing a comparative framework. Therefore, solutions in blockchain architectures that address such challenges as scalability and interoperability gaps, workload balancing and efficient use of computational overhead are required. Practice implications are for the use of hybrid blockchain-IoT models for critical infrastructure and standardization of regulations to allow secure integration. Future work includes research on deployment in the real world, long-scalable studies, and integration of real-world data into emerging technologies, such as zero-trust environments. However, the limitations of the study, like theoretical models being used while the data is limited to the industry itself, suggest a larger empirical validation body. In summary, this paper advocates the use of blockchain for IoT security, as it offers and represents the transformative potential for IoT security, but relief should be given to the challenges associated with its implementation.

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